

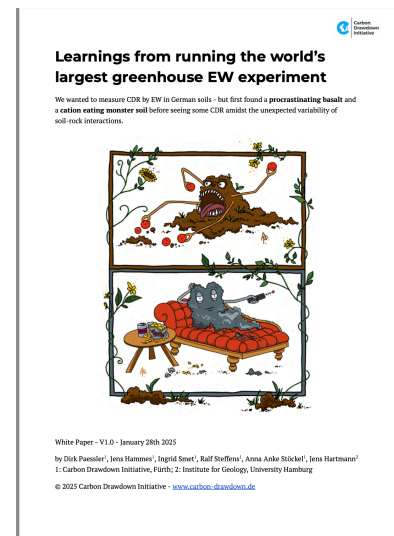


What We Learned from the World's Largest Greenhouse Experiment

January 29, 2025

THE UNEXPECTED CHALLENGES OF ENHANCED WEATHERING

Enhanced Weathering (EW) has long been considered a promising climate solution, leveraging natural rock dissolution to remove CO₂ from the atmosphere. The concept is simple: apply finely ground silicate rocks to soils, let them dissolve, and capture carbon in the form of stable bicarbonate. But as we have learned, real-world soil-rock interactions are anything but straightforward.



Over the past four years, we conducted the world's largest **greenhouse EW experiment**, aiming to measure carbon dioxide removal (CDR) across different soil-rock combinations (all soils came from Germany). Instead of clear-cut CDR effects, we first encountered a “**procrastinating basalt**”—a rock that barely contributed to measurable

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Our dataset, built from **6,500 water samples, 4,500 manual titrations, and 2,000 ICP-MS runs**, revealed a striking conclusion: **not all soil-rock combinations lead to meaningful alkalinity-based CDR effects within relevant timeframes**. In fact, more than half of our tested variations failed to show a statistically significant increase in alkalinity-based carbon removal over two years, a critical window for scaling EW as a viable climate solution.

However, our experiment also uncovered successful soil-rock pairings. **Steel slag**, an industrial byproduct, emerged as the most effective amendment, achieving CDR rates of up to **2.8 tCO₂ per hectare per year**. This reinforces the importance of **dissolution kinetics and material surface area** in determining the effectiveness of rock amendments for carbon sequestration.

These findings have major implications for the future of EW. Instead of relying solely on theoretical weathering rates, we need **big data approaches and machine learning models** to predict the best-performing soil-rock combinations. The climate crisis does not allow us the luxury of decades-long trial-and-error approaches—**we must accelerate the identification of effective EW strategies now**.

Our study raises fundamental questions for the EW

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In 2025 we will extent our experiments using two strategies:

- As we are currently dismantling the experiment we are taking a load of soil samples which will be processed to find clues on what happened in the soil, have the cations been lost to clay formation or will they be helpful with CDR some time in the future.
- We are going to set up an even more ambitious follow-up experiment.

The full white paper provides a detailed analysis of our findings, challenges, and next steps. Read on to explore the insights shaping the future of Enhanced Weathering.

ABSTRACT

Here we present in-depth monitoring results of up to 40 monthly chemical and physical soil parameters representing several hundred long-term Enhanced Weathering (EW) pot experiments conducted over the last four years, based on 6,500 water samples, 4,500 manual titrations and 2,000 ICP-MS runs. We examined 14 amendments including basalt, dunite, meta-basalt, limestone, glacial sediment, and steel slag, and tested these inputs within a diverse set of 17 agricultural soils from Germany.

3 years ago when planning these experiments we had defined

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effects for more than 40 soil/rock combinations based on data from 4 replicates.

We found that soils have a significant influence on CDR effects. Over a 2-year period only 16 out of 49 variations (32%) in our greenhouse showed an increase-in-leachate-alkalinity-based carbon removal rate of at least 0.5 tCO₂/ha/year or more. Only 26 variations (54%) showed statistically significant evidence of CDR over the period of 2 years, a critical time frame in the context of rapidly scaling up EW as a CDR technology.

It is thereby unfortunate that the soil we used most extensively throughout our experiments (collected from our field in Fürth) turned out to be a bad choice as it seems to nearly always retain the weathering products (“cation eating soil”), at least over the duration of our observations of 2-3 years. Likewise, our most tested amendment (a German basalt rock dust) showed little to no increase in leachate alkalinity (“procrastinating basalt”) during our experiment. Especially the combination of this Fürth soil and basalt, which we tested in many variations, gave many disappointing results.

While we did find some combinations of other soils and rocks that resulted in measurable CDR, the overall increase in titration alkalinity of the leachate was lower than we

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projects in Germany, or even indicative for temperate mid-latitudes in general.

Additional solid phase laboratory data should soon help us to better understand the reasons for the observed behaviour (e.g. cations trapped in CEC, MAOM, or lost in clays), as soil samples will be taken during the dismantling of the pots in early 2025. Without this data we cannot explain the results fully and can only report our present learnings.

As for the soil/rock combinations that did result in a significant increase in titration alkalinity in the leachate compared to control, the one amendment that outshone all others and showed a CDR effect on the 3 main soils was the steel slag (depending on the soil, between 1.0 and 2.8 tCO₂/ha/year). This likely reflects the combination of its high specific surface area and very quickly dissolving Ca-Minerals (20% portlandite, 8% calcite). The other industrial material was tested only on the soil yielding the lower CDR rates for steel slag (clay-rich soil with neutral pH), yet achieved a CDR rate of 2.6 tCO₂/ha/year. Runners-up with high CDR rates within the first year of application on a more acid and silty soil are limestone (0.9 tCO₂/ha/year) and glacial sediment (1 tCO₂/ha/year). Whereas the performance of the first is likely due to its quickly dissolving calcite content (85%), in case of the glacial sediment with more slowly dissolving silicate

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tCO₂/ha/year on all three soils. These results seem to suggest that the amendment properties most influencing CDR effects are the dissolution kinetics of the main minerals and the material's specific surface area.

Our main conclusion from the above observations is that a follow-up experiment with many more soil-rock variations and solid phase data, as well as leachate measurements, is necessary to further explore EW process results, to find out where the cations actually go, and to build a database which could be used to develop/test data-based MRV models for EW.

Strategic consequences for the scale-up of EW as a potential CDR solution

We believe our results support our earlier suggestion to move towards a big data approach using machine learning for the EW industry that mostly considers the complex processes (i.e. “how it works?”) in the soil as a black box, because it may very well take too long to figure that all out. Even though eventually we do need a deeper understanding of the EW soil processes, the climate crisis does not give us the necessary decade(s) for this. So, whilst continuing to work on detailed understanding, we think a major part of EW research efforts should be put into building many more experiments in order to create a large data set on which numerical models can be built. We believe that a large number of pot/greenhouse

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[ERW Data Quarry](#) initiative.

DOWNLOAD THE PDF

The [white paper](#) is available as PDF ([41 pages](#), [10 MB](#), [pre-print](#)).

DOI (pre-print): <http://dx.doi.org/10.13140/RG.2.2.18184.74247>



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